

Clinic Appointment Application

Problem Statement and Project Scope

Patients often need to contact clinics by phone or visit in person to identify an appropriate doctor and request an appointment. This manual process can make doctor information difficult to compare, appointment details harder to track, and scheduling conflicts more likely when availability is not checked consistently.

The proposed Clinic Appointment Application provides a mobile interface for browsing a doctor directory and requesting, reviewing, editing, or cancelling appointments. A Node.js and Express API manages authentication, doctor and appointment records, and a MongoDB Atlas database stores the application data.

Project Aim

Design and implement a small full-stack mobile application that lets authenticated users manage a doctor directory and book appointments while preventing overlapping active bookings for the same doctor.

Objectives

- Provide account registration and login with bcrypt password hashing and JWT authentication.
- Support doctor create, read, update, and delete operations, including a validated profile image upload.
- Support appointment create, read, update, and delete/cancel operations linked to a user and a doctor.
- Reject invalid, past, or overlapping appointment time ranges.
- Deploy the backend against MongoDB Atlas and configure the mobile client to use the hosted API.

Core Entities and Business Rule

Doctor (primary): name, specialty, contact details, description, and image URL. **Appointment (related):** user reference, doctor reference, start and end times, reason, and status (Pending, Confirmed, or Cancelled). User records support authentication and are not counted as one of the two system entities.

A requested interval is rejected when the same doctor already has a non-cancelled appointment with an interval that overlaps it. Cancelled appointments do not block availability. The check is repeated when an appointment is edited, excluding the appointment being edited.

Scope and Limitations

The application focuses on authentication, two domain entities, image upload, appointment status, and the overlap rule. Payments, email/SMS delivery, real-time features, analytics, and a separate web administration portal are outside the intended scope. Doctor-management endpoints are protected by login; a separate administrator role is not currently part of the implemented design.

Topic approval: Lecturer approval status must be confirmed before submission. Backend deployment URL: TBD.